Soft Robotic Assistant

Literature-led requirements and architecture study for a compliant upper body

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13 November 2026

Engineering design report. Evidence: literature synthesis and transparent calculations; no prototype, simulation campaign, human-subject experiment or clinical validation.

Abstract

A freestanding robot intended to guide or steady a person must satisfy two different demands: apply useful assistance and remain predictable when a person applies a much larger unplanned load. This report documents the evidence and requirements work for a compliant, two-arm assistive upper body. The retained repository contains 205 annotated source entries and 287 bibliography records. Its principal output is a traceable distinction between reported capabilities, inferred design quantities and unresolved safety questions. A 28 N end-effector target is a selected planning assumption; shoulder and elbow gravity-compensation values of approximately 10 N m and 3 N m depend on inferred moment arms. Published hugging and inflatable robots defeat a broad claim of architectural novelty. Simple statics instead motivate a grounded load path, independently measured contact forces and a defined low-force task. These are design directions, not demonstrated capabilities. The report derives the relevant screening equations, explains why industrial contact thresholds cannot be transferred directly to frail users, and specifies the evidence needed before hardware or interaction studies could support stronger claims.

Completed scope. This report closes a literature-led requirements and analytical feasibility study at commit `f3f3d3764ca2`. It preserves the 205 annotation rows, the force/load-path assumptions and the proposed verification programme. No robot, controller implementation, FEA campaign or human-subject experiment forms part of this completed scope. Further device development would be a separately specified project.

1 Problem and individual contribution

Central question. What force, load-path and contact evidence is needed to turn a soft upper-body concept into a measurable low-force assistance task?

Current answer. The literature supports a task-specific bench specification and explicit distinction between active assistance, gravity support and accidental reaction loads. It does not establish a working robot or a safe envelope for frail users. The candidate hypothesis for a future instrumented bench study is that a grounded load path and compliant transmission can provide the selected force while preserving defined release behavior; it remains untested.

The starting brief was a soft, freestanding humanoid upper body for physical interaction with elderly and rehabilitation users. A torso and two arms can suggest reassuring contact, but a soft exterior says little about whether an arm can support a sustained load, release safely during a fault or avoid tipping. The useful engineering question is therefore narrower: *what low-force task can be specified, instrumented and characterized without confusing assistance capability with user safety?*

Mergen Ulziibayar's documented contribution is the literature-program structure, source annotation and grading, derived requirements, prior-art and patent search, and the evidence-boundary record. The repository is an evidence and design brief, rather than a completed robot. AI-assisted tools supported source organization, checking and preparation of this report; accountability for the selected assumptions

and future design decisions remains with the project owner. Neither an automated summary nor a literature citation is a substitute for reviewing a source's conditions of use.

The public record examined here is commit `f3f3d3764ca2` of the project repository [1]. Its annotation matrix and requirements record are the basis for the census and numerical assumptions below. This report introduces no new hardware results and does not turn the design brief into a patient study.

2 Literature method and quality of evidence

2.1 An annotation is a design input, not a vote

The matrix records a source's citation, verification route, contribution, numerical information, design relevance, evidence grade and limitations. Its 205 rows cover 14 repository sections; 287 BibTeX records include a wider reference set and duplicate or contextual records. These counts describe the retained documentation. They are not 205 independent experiments, a systematic-review completeness claim or a quality score.

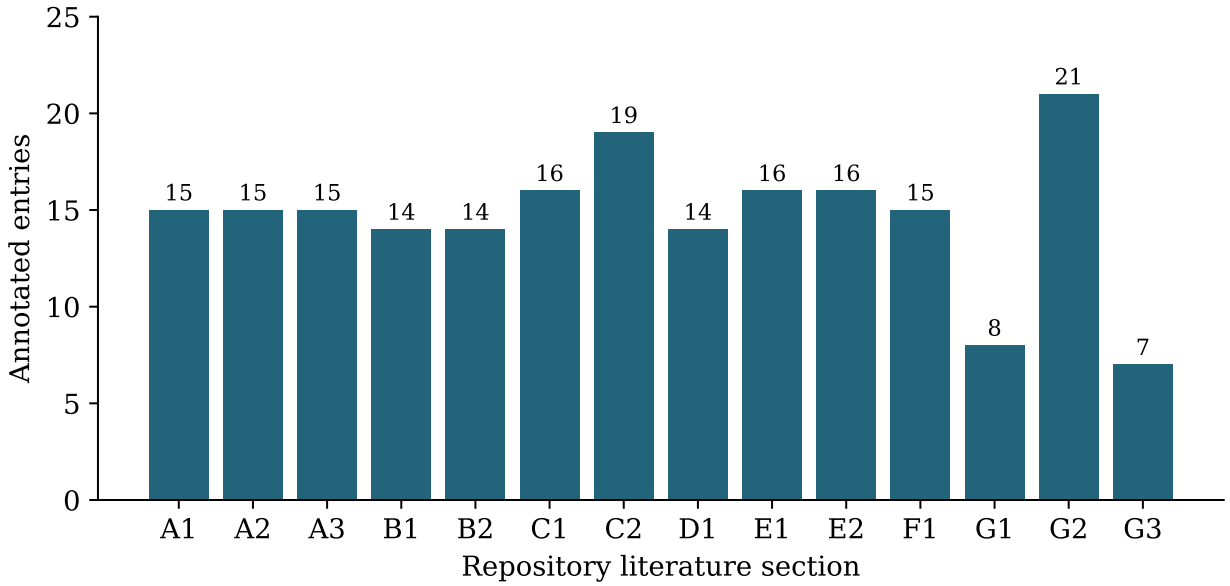


Figure 1: Census of the retained 205 annotation rows, grouped by their repository section identifiers. Replotted directly from the source CSV for this report. Groups: A actuation; B structures/sensing; C safety/testing; D control; E rehabilitation/interaction; F comparables; G patents, citation and standards verification. Counts show coverage, not independent replications or evidence strength. Source: Ulziibayar, project literature matrix [1].

The source distinguishes human/hardware or standards evidence, bench hardware, simulation/concept work, reviews, and industry claims. These labels are useful only when read with the recorded limitation: a bench force measurement may be strong evidence of a particular actuator's blocked force and weak evidence of an assembled arm's sustained assistance. An industry payload claim is not a controlled contact study. A human study on healthy adults is not an injury threshold for frail users.

The review process also corrected attractive but unsupported statements. RI-MAN's cited demonstration involved a doll rather than a person. Assumed torque values had to be separated from measured assistance. Some source metadata and publication labels required correction. These corrections are a material output: they prevent a plausible-looking number from becoming a requirement through repetition [1].

2.2 Nearest prior art and the claim that remains

HuggieBot is a close reference for human-sized, freestanding, soft-surfaced interaction. Block et al. describe an inflated sensing torso, articulated arms and studies with 32 and 16 users across the reported iterations [2]. These studies concern hugging interactions and responsive behavior; they do not validate this project's force target or rehabilitation purpose. Niiyama et al. demonstrate blower-powered inflatable joints with internal tendon actuation and a multi-joint hugging example [3]. Together, these precedents make "a soft humanoid that interacts physically" an inadequate novelty claim.

The repository also records King Louie, pneumatic fascicles, wearable assistance, compliant sensing and patient-handling machines. The relevant comparison is not a single payload leaderboard. A wearable device routes reaction loads through the wearer; a freestanding arm routes them through its torso and base. A low-pressure inflatable structure and a rigid arm beneath padding have different fault modes even if both are described as soft. A literature search that finds no measured envelope for a particular task does not prove that no such system exists anywhere.

The defensible future objective is an *instrumented contact-force envelope for a specified guiding, steadying or positioning task*. It would report force, pose, speed, duration, support conditions and fault response together. That objective is compatible with prior art and can be tested; it has not yet been achieved here.

3 Requirements and assumptions

Table 1 keeps the main planning quantities separate from their evidence status. An assistive-force selection is not a safety limit, a structural load is not an operating command, and an inferred gravity torque is not an actuator measurement.

Table 1: Planning quantities retained from the source brief [1].

Quantity	Planning value	Interpretation and unresolved input
End-effector assistance	28 N	Selected task assumption informed by rehabilitation literature; not a patient-safe ceiling.
Contextual force ceiling	45 N	Literature comparison retained in the brief; does not authorize applying this force to a person.
Assist-as-needed variation	approximately ± 6 N	Narrow source context; the intended task and control law still require specification.
Shoulder gravity torque	approximately 10 N m	Inferred with a 0.30 m center-of-mass arm; anthropometry and pose are unresolved.
Elbow gravity torque	approximately 3 N m	Inferred with a 0.16 m moment arm; not a measurement of an assembled limb.
Structural reaction	at least 330 N	A separate lean/support planning case at hand height; base geometry and load direction are unresolved.
Impact acquisition	at least 10 kHz	Proposed instrumentation rate for millisecond events; sensor bandwidth, calibration and filtering remain to be established.

The source requirements use a rehabilitation device and selected assist-as-needed studies as context [4, 5]. That provides a starting magnitude, not a universally appropriate prescription. Whether the robot is intended to touch, guide, support body weight or deliver therapy changes its risk analysis, regulatory classification, mechanical requirements and ethics review. This choice precedes detailed optimization.

Several inputs cannot be chosen by averaging papers: intended users, contact body region, posture, supervision, allowable support function, reachable workspace, load duration and behavior on loss of

power. The design must also distinguish passive support from active assistance. A device that is easily backdriven can release unwanted force but may collapse under a sustained lean; a device that holds without power can trap or constrain a person unless release is designed explicitly.

4 Mechanical screening calculations

4.1 Force and joint torque are different quantities

For a force $\mathbf{F}$ applied at position $\mathbf{r}$ relative to a joint,

$$\tau = \mathbf{r} \times \mathbf{F}, \qquad |\tau| = F r_{\perp}. \tag{1}$$

The 28 N target produces 8.4 N m at a perpendicular 0.30 m reach and 14 N m at 0.50 m. These are illustrative arithmetic checks using the selected force, not measured arm performance. Joint torque also includes limb self-weight, inertia, friction and any external load. A torque specification cannot be converted to end-effector assistance without configuration and lever-arm information.

For a supported segment of mass m and center-of-mass offset r_c, the maximum gravity moment in a horizontal pose is

$$\tau_g = m g r_c \sin\theta, \qquad \tau_{g,\max} = m g r_c, \tag{2}$$

where θ is the angle between gravity and the position vector. The source's 10 N m at 0.30 m implies an effective supported mass of about 3.4 kg; 3 N m at 0.16 m implies about 1.9 kg. These are *implied masses*, not an independently validated anthropometric model. If the limb load is distributed across two segments, a shoulder calculation must sum each segment's contribution rather than substitute a single forearm mass.

Moment-arm uncertainty matters directly. Holding mass constant, changing the assumed shoulder COM arm from 0.30 m by 0.05 m changes the predicted gravity torque by about 17%. The same 0.05 m error around the elbow assumption is about 31%. This sensitivity alone prevents the two rounded torque figures from serving as final actuator-sizing values.

4.2 Grounding and tipping

Useful assistance requires a closed reaction path. In a simplified planar free-body diagram, a horizontal hand force F_h at height h creates an overturning moment $F_h h$. If the base weight is W and its center of mass is b inside the tipping edge, a static margin can be written

$$M_{\text{margin}} = W b - F_h h - M_{\text{other}}, \qquad M_{\text{margin}} > 0. \tag{3}$$

Here M_{other} includes arm self-weight and vertical or off-axis applied forces. This expression neglects dynamic impact, caster motion, floor slope, sliding and compliance. It is a screening relation, not a qualified tipping test.

An illustrative 330 N horizontal reaction at 1 m, resisted with a 0.30 m support arm, would require more than 1100 N of weight before any additional margin. That is approximately 112 kg equivalent mass. No such base is selected. The example shows why a low assistance target does not automatically imply a light, freely standing support robot. Sliding must be checked separately, for example against $F_h \le \mu W$ under an explicitly measured floor contact condition.

4.3 Inflatable structure screen

The brief uses an idealized cylindrical inflatable bending relation

$$M_{\text{screen}} = \frac{P \pi D^3}{8}. \tag{4}$$

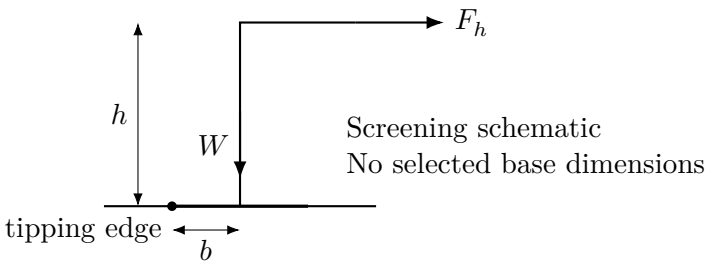


Figure 2: The base must resist the external moment, independently of whether the arm's contact surface feels compliant. Diagram derived for this report from the static load-path requirement; not a CAD design.

At 20 kPa and diameter 0.15 m, it gives about 26.5 N m. A 200 N lean at 0.50 m creates 100 N m. This comparison explains concern about routing a large support moment entirely through a small compliant torso. Equation 4 is an idealized screening assumption in the brief: seam geometry, fabric anisotropy, wrinkling, pressure stability, joints and fixture conditions are absent. It must not be presented as a certified failure pressure or the capacity of an unbuilt structure.

Because the expression scales as PD^3, diameter is influential; pressure alone is not a free solution. Increasing pressure changes the contact behavior and stored energy, while increasing diameter changes workspace and human access. A grounded structure covered by compliant material is therefore a candidate architecture, not a contradiction of the soft-contact goal.

4.4 Reconciliation of the imported velocity screen

The source brief places a 1.06 m/s chest-governed velocity beside a later approximately 110 N blunt-impact force selection. Those two values use different force bases. A consistent illustrative spring/impact calculation uses the relative-speed relation retained in the literature notes,

$$\mu = \left(m_H^{-1} + m_R^{-1}\right)^{-1}, \qquad v_{\text{screen}} = \frac{F_t}{\sqrt{\mu k}}. \tag{5}$$

Here m_R is effective translational robot mass, not a rotational moment of inertia. Keeping the notes' assumed chest effective mass $m_H = 40\,\text{kg}$, $m_R = 3\,\text{kg}$ and stiffness $k = 25\,\text{N/mm} = 25\,000\,\text{N/m}$ gives $\mu = 2.790\,70\,\text{kg}$. The force F_t is already the transient force used in this equation; no second factor of two is applied to it.

Table 2: Independent arithmetic reconciliation at fixed mass/stiffness assumptions. Inputs are retained literature-screening values, not measurements of this device or clinical limits.

Force basis	F_t (N)	v_{screen} (m/s)
Earlier transient-force basis	280	1.0601
Later approximately sternum-force basis	110	0.4165

The approximately 0.42 m/s result is the lower conditional screen after the force substitution. It does not establish acceptable contact with elderly or rehabilitation users. The notes' effective masses, contact area and stiffness are not device measurements; the industrial standard's applicability and the population-specific force basis remain separate questions [1, 8, 7]. The earlier 1.06 m/s value must therefore not remain a governing device-speed claim alongside 110 N.

5 Architecture implications

5.1 Separate the structural path from the interaction layer

A candidate system can use a stable load-bearing skeleton, compliant transmission and replaceable soft covers. Each has a different purpose: the skeleton closes gravity and support loads; transmission compliance limits mechanical impedance over a defined range; the cover spreads local contact and prevents exposed edges. None removes the need for pinch-point analysis or measured force control. A rigid mechanism under foam is still capable of trapping a limb.

The actuator comparison must include sustained static torque, temperature, stiffness, pressure/electrical supply, stroke, reversibility and release behavior. Peak specific power or a blocked-force result from a fascicle does not establish continuous whole-arm assistance [6]. Pneumatic systems bring hose routing, valve leakage, pressure monitoring and depressurization behavior; electric systems bring drive inertia, gear friction and thermal stall. These are architecture tradeoffs, not a ranking from the present evidence.

5.2 Compliance, control and support loss

In a simple series elastic transmission,

$$\tau_s = k_s \left(\theta_m / N - \theta_j \right), \tag{6}$$

where k_s is spring stiffness and N is gear ratio. Deflection can provide a torque estimate only if spring calibration, hysteresis, friction and sensor offsets are characterized. A controller may regulate impedance or admittance, but stability depends on the complete human-robot dynamics, sample delay, contact stiffness and saturation. The brief's discussion of controller families is not evidence that one controller is safe in all contacts.

Gravity compensation and an explicit support-loss state should be established before ambitious interaction behavior. A brake, passive support or controlled release might each be appropriate in a different task. Their transitions need defined observable conditions: commanded stop, power failure, pressure loss, sensor disagreement and emergency release. A brake that simply locks can preserve a load while preventing withdrawal, so "holds safely" and "releases safely" must be tested separately.

5.3 Instrument the quantity that matters

Tactile skin can identify contact location, but contact intensity and load-path reaction need calibrated force sensing. A proposed bench configuration should measure commanded force, end-effector reaction, base reaction, pose and time on a synchronized clock. Chamber pressure or motor current alone is not a verified whole-body contact-force measurement. Elastic hysteresis, leaks, cable friction and posture can map the same actuator reading to different external forces.

The proposed 10 kHz sample rate gives 60–100 nominal samples across a 6–10 ms event. It does not guarantee that the transducer or conditioning electronics resolves that event. A 100 Hz low-pass filter can change a peak substantially; the exact filter order, phase behavior, raw record and reported peak definition must accompany the result. A force threshold without acquisition bandwidth and calibration uncertainty is incomplete.

6 Safety scope and evidence boundaries

The official ISO scope pages distinguish personal-care robots from industrial collaboration and from robots regulated as medical devices [7, 8]. ISO/TS 15066:2016 explicitly concerns industrial systems; its principles may inform other robotics, but its applicability is not established for this project. The ISO 13482:2014 scope includes physical assistants and notes the lack of exhaustive internationally recognized

impact injury data. The cited editions and scope pages were checked for this report; a design review must use the edition applicable when that review occurs.

Consequently, the brief's imported industrial velocity, pressure and impact numbers are *literature-screening context*. This report deliberately does not promote them into patient-safe operating limits. Frail users, anticoagulant medication, reduced sensation, vulnerable skin and supported postures can make a pain-based value inappropriate. No numerical derating factor is justified by this evidence set. Selection of a target population and qualified risk/clinical review are necessary before claiming an interaction envelope is suitable for people.

The most important hazards are not restricted to impact. A slowly moving mechanism can clamp or shear; a compliant structure can buckle; a robot can lose support or tip; a brake can prevent withdrawal; a pneumatic leak can alter stiffness while the controller remains operational. A visible soft surface does not establish a safe state after a single fault.

The patent search is a record of prior art, not a freedom-to-operate opinion. The repository identifies pressure-only estimation claims relevant to a sensing concept. Their geographic scope, claim interpretation, validity and applicability require specialist review before commercial decisions. No assertion here establishes infringement or clearance.

7 Verification programme proposed by the evidence

The evidence supports a staged bench programme rather than immediate human trials. Table 3 states what a future measurement would actually establish. It is a proposed plan, not completed testing or authorization to begin physical work.

Table 3: Evidence needed to turn the brief into a characterized device.

Question	Proposed measurement	What must be reported
Static assistance	Instrumented load at representative poses and reach	Applied force, joint angles, duration, temperature, drift and repeatability; no human surrogate for calibration.
Structural support	Defined load directions at hand height with restrained test environment	Base reactions, deflection, sliding/tipping tendency, support footprint and unloading behavior.
Release behavior	Power, pressure and sensing fault injection on a bench fixture	Time to force reduction, residual trapped force, brake state and manual release access.
Contact distribution	Calibrated interface or qualified surrogate under specified contact shapes	Force and pressure distributions, raw/filtered bandwidth and sensor uncertainty.
Dynamic behavior	Controlled trajectories and contact stiffness changes	Tracking error, overshoot, delay, saturation and stability domain under the tested conditions.
Endurance	Repeated, defined duty cycle after safe bench operation	Load, cycle shape, hold time, temperature, leaks, wear and changes in force calibration.

Each experiment needs an acceptance condition tied to its purpose before data are collected. A static force test cannot establish clinical benefit; a passing tipping case cannot establish every off-axis condition;

an intact prototype after repeated motion does not characterize contact injury. The force target may change once task feasibility and qualified safety review are available.

8 Uncertainty and limitations

The largest uncertainty is the task definition, followed by anthropometry/load sharing, contact behavior and architecture-dependent fault response. Calculation precision is a minor issue compared with these uncertainties. A final actuator choice requires a torque-speed-duty envelope and measured transmission behavior. A final base requires footprint, center of mass and external load directions. A contact claim requires appropriate human/surrogate evidence and review.

The present work nevertheless changes the design problem concretely. It replaces a broad novelty claim with a testable measurement objective, labels the 28 N target as a choice, exposes the inferred basis of two gravity torques, and separates useful assistance from much larger structural reactions. Published precedent supports the feasibility of soft-surfaced and inflatable interaction systems; it does not establish that this proposed device works. The appropriate next output is a task-specific, instrumented bench concept with explicit support and release behavior.

9 Conclusions

1. The completed contribution is a literature/requirements study, supported by 205 annotation rows and a traceable planning table (Fig. 1, Table 1); it does not demonstrate an assistive device.

2. The 28 N assistance target is a task choice. The approximately 10 N m shoulder and 3 N m elbow gravity torques depend on inferred moment arms and load distribution (Eqs. 1–2). They cannot be used as measured actuator capability.

3. Base stability is a separate load-path problem: the illustrative 330 N horizontal reaction at 1 m requires more than 1100 N of resisting weight with a 0.30 m support arm before additional margin (Eq. 3, Fig. 2).

4. With unchanged assumed contact stiffness and effective masses, substituting 110 N for 280 N reduces the conditional impact-model velocity from 1.060 m/s to 0.416 m/s (Eq. 5, Table 2). The lower value governs that arithmetic screen; neither value is a patient-safe operating limit.

5. A task-specific grounded load path, measured external force and explicit release/support-loss behavior are the appropriate design direction (Table 3). Detailed hardware selection should follow those requirements rather than a general claim that compliance establishes safety.

10 Future work

1. Begin with the cheapest falsifying bench experiment: a restrained surrogate load path with a calibrated force transducer and known applied load, measuring pose, reaction, deflection and unloading. Define the intended task and review the protocol before physical work.

2. Characterize one candidate transmission's effective mass, compliance, friction and sustained torque/temperature across declared poses; use the measurements to revise the analytical envelope.

3. Seek the applicable risk, ethics and clinical review before any human-contact study. Patient suitability requires evidence specific to the intended population and task.

A Reproducibility and source map

Record	Use in this report
`README.md`	Project scope and retained evidence count.
`literature/literature_matrix.csv`	205 annotation rows; Figure 1 is a direct count by section.
`literature/refs.bib`	Bibliographic metadata; counts do not imply unique independent experiments.
`docs/requirements.md`	Selected force, inferred torque, structural and acquisition assumptions.
`docs/prior-art.md`	Architectural comparisons; broad absence claims are narrowed here to this corpus.
`docs/evidence-limits.md`	Source corrections, applicability limits and unresolved verification.

The accompanying machine-readable source record stores full commit identity, exact source paths, SHA-256 values and claim-to-source mappings. Figure 1 was produced by counting existing rows, not by adding experiments or reclassifying source quality. All diagrams in this report are original explanatory graphics; no third-party robot photograph is reproduced. Repository source text is attributed to Mergen Ulziibayar; external research is cited rather than copied at length.

B Minimum specification before component sizing

Before procurement or a detailed mechanical drawing, the design brief should state: one intended task; user posture and supervision; reachable contact regions; force and speed commands; worst credible support reaction; sustained load duration; release and support-loss states; pressure or electrical supply envelope; and who reviews interaction risk. These are missing design inputs rather than numerical performance failures. A completed specification would allow the requirements table to become testable without implying approval of patient contact.

References

[1] M. Ulziibayar. *Soft Robotic Assistant for Physical Interaction with Elderly and Rehabilitation Patients*. Source repository, commit `f3f3d3764ca2bfcfe5a1a26b466d771a885d87f9`. `https://github.com/500ft/Soft-Robotic-Assistant-for-Physical-Interaction-with-Elderly-and-Rehabilitation-Patients/tree/f3f3d3764ca2bfcfe5a1a26b466d771a885d87f9`.

[2] A. E. Block, H. Seifi, O. Hilliges, R. Gassert and K. J. Kuchenbecker. In the Arms of a Robot: Designing Autonomous Hugging Robots with Intra-Hug Gestures. *ACM Transactions on Human-Robot Interaction*, 12(2), 2023. `https://doi.org/10.1145/3526110`. Author manuscript: `https://arxiv.org/abs/2202.09935`.

[3] R. Niiyama, Y. A. Seong, Y. Kawahara and Y. Kuniyoshi. Blower-Powered Soft Inflatable Joints for Physical Human-Robot Interaction. *Frontiers in Robotics and AI*, 8:720683, 2021. `https://doi.org/10.3389/frobt.2021.720683`.

[4] H. I. Krebs, N. Hogan, M. L. Aisen and B. T. Volpe. Robot-aided neurorehabilitation. *IEEE Transactions on Rehabilitation Engineering*, 6(1):75–87, 1998. `https://doi.org/10.1109/86.662623`. Numerical selections here follow the project's annotated requirements record, not a new clinical interpretation.

[5] A. P. Arantes et al. Evaluation of a novel real-time adaptive assist-as-needed controller for robot-assisted upper extremity rehabilitation following stroke. *PLOS ONE*, 18(10):e0292627, 2023. `https://doi.org/10.1371/journal.pone.0292627`.

[6] M. A. Robertson, H. Sadeghi, J. M. Florez and J. Paik. Soft Pneumatic Actuator Fascicles for High Force and Reliability. *Soft Robotics*, 4(1):23–32, 2017. `https://doi.org/10.1089/soro.2016.0029`. Used as bench-actuator context in the reviewed corpus.

[7] International Organization for Standardization. *ISO 13482:2014: Robots and robotic devices—Safety requirements for personal care robots*. Official scope page, consulted 11 November 2026. `https://www.iso.org/standard/53820.html`.

[8] International Organization for Standardization. *ISO/TS 15066:2016: Robots and robotic devices—Collaborative robots*. Official scope page, consulted 11 November 2026. `https://www.iso.org/standard/62996.html`.